

Zebin Xing

Institute of Automation, Chinese Academy of Sciences, Beijing

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Education

Institute of Automation, Chinese Academy of Sciences (CASIA)

Academic Master in Computer Application Technology | Advisor: **Prof. Qichao Zhang**

Beijing, China

Sept 2024 - Jun 2027

- GPA: 3.6 / 4.0

Beijing University of Posts and Telecommunications (BUPT)

Bachelor in Software Engineering (National First-class Discipline Program)

Beijing, China

Sept 2020 - Jun 2024

- GPA: 3.8 / 4.0 (**top 5%**)

Awards

Awards during Undergraduate and Postgraduate Studies

- Merit Student, University of Chinese Academy of Sciences, 2025.
- Beijing Outstanding Graduate, (**top 5%**), Beijing Municipal Education Commission, 2024.
- First-class Scholarship, Beijing University of Posts and Telecommunications, 2022.
- First Prize of the Chinese Mathematics Competitions for Undergraduates, Chinese Mathematical Society, 2022.
- National Endeavor Scholarship, Ministry of Education of the PRC, 2021.

Research Interest

My **long-term research goal** is to build Physical AI agents that can perceive, predict, and interact with the real world in a reliable and generalizable way.

My research focuses on world models, vision-language-action models, and multimodal planning, with the aim of transforming raw sensory observations into grounded physical reasoning for safe decision-making across robotic systems and autonomous driving.

Research Experience

AnyVerse (无界动力)

Research Intern @ **AnyVerse**

Beijing, China

May, 2026 - Now

- Topic:** Visual-Tactile Fusion; World Model.

CHANGAN Automobile (长安汽车)

Research Intern @ **DISCOVER Lab** | Advisor: **Dr. Qiang Chen**

Beijing, China

Nov. 2025 - Apr. 2026

- Topic:** Vision-Language-Action model; E2E Autonomous Driving; Memory.

Horizon Robotics (地平线)

Research Intern @ **Horizon Robotics** | Advisor: **Dr. Wei Yin**

Beijing, China

Jun. 2024 - Apr. 2024

- Topic:** E2E Autonomous Driving, Flow Matching Policy, World Model.

Wireless Communication Security Center, BUPT (北邮无线通信安全中心)

Research Intern @ **Wireless Communication Security Center** | Advisor: **Prof. Guoshun Nan**

Beijing, China

Sept. 2022 - Feb. 2023

- Topic:** AI safety; Differential Privacy; Federated Learning.

Publications

(* denotes equal contribution; † denotes corresponding author.)

FIRST / CO-FIRST AUTHOR PUBLICATIONS

DriveVLA-M0: Failure-Aware Memory Augmentation for Autonomous Driving

Zebin Xing, Yupeng Zheng, Qiang Chen, Linbo Wang, Yichen Zhang, Pengxuan Yang, Junli Wang, Deheng Qian, Xiaoqing Ye, Junyu Han, Yifeng Pan, Qichao Zhang†, Dongbin Zhao

ACM MM, 2026

Mimir: Hierarchical Goal-Driven Diffusion with Uncertainty Propagation for End-to-End Autonomous Driving

Zebin Xing, Yupeng Zheng, Qichao Zhang†, Zhixing Ding, Pengxuan Yang, Songen Gu, Zhongpu Xia, Dongbin Zhao

IEEE Robotics and Automation Letters (RA-L), 2025

TrajMoE: Scene-Adaptive Trajectory Planning with Mixture of Experts and Reinforcement Learning

Zebin Xing, Pengxuan Yang, Linbo Wang, Yichen Zhang, Yiming Hu, Yupeng Zheng, Junli Wang, Yinfeng Gao, Guang Li, Kun Ma, Long Cheng, Zhongpu Xia, Qichao Zhang†, Hangjun Ye, Dongbin Zhao

International Annual Conference on Complex Systems and Intelligent Science (CSIS-IAC, Best Paper), 2026

GoalFlow: Goal-Driven Flow Matching for Multimodal Trajectories Generation in End-to-End Autonomous Driving

Zebin Xing*, Xingyu Zhang*, Yang Hu, Bo Jiang, Tong He, Qian Zhang, Xiaoxiao Long, Wei Yin†

CVPR, 2025

PlanAgent: A multi-modal large language agent for closed-loop vehicle motion planning

Yupeng Zheng*, **Zebin Xing***, Qichao Zhang†, Bu Jin, Pengfei Li, Yuhang Zheng, Zhongpu Xia, Kun Zhan, Xianpeng Lang, Yaran Chen, Dongbin Zhao
IEEE Transactions on Cognitive and Developmental Systems (TCDS), 2025

OTHER PUBLICATIONS

PokéVLA: Empowering Pocket-Sized Vision-Language-Action Model with Comprehensive World Knowledge Guidance

Yupeng Zheng*, Xiang Li*, Songen Gu*, Shuai Tian, Weize Li, Lingbo Wang, Senyu Fei, Pengfei Li, Yinfeng Gao, **Zebin Xing**, Yilun Chen, Qichao Zhang, Haoran Li†, Wenchao Ding†
IEEE Transactions on Robotics (T-RO), *In Submission*, 2026

Latent-WAM: Latent World Action Modeling for End-to-End Autonomous Driving

Linbo Wang, Yupeng Zheng, Qiang Chen, Shiwei Li, Yichen Zhang, **Zebin Xing**, Qichao Zhang†, Xiang Li, Deheng Qian, Pengxuan Yang, Yihang Dong, Ce Hao, Xiaoqing Ye, Junyu Han, Yifeng Pan, Dongbin Zhao
ACM MM, 2026

DreamerAD: Efficient Reinforcement Learning via Latent World Model for Autonomous Driving

Pengxuan Yang, Yupeng Zheng, Deheng Qian, **Zebin Xing**, Qichao Zhang†, Linbo Wang, Yichen Zhang, Shaoyu Guo, Zhongpu Xia, Qiang Chen, Junyu Han, Lingyun Xu, Yifeng Pan, Dongbin Zhao
ECCV, 2026

Beyond Imitation: Learning Safe End-to-End Autonomous Driving from Hard Negatives

Junli Wang, Zhihua Hua, Xueyi Liu, **Zebin Xing**, Haochen Tian, Kun Ma, Hangjun Ye, Guang Chen, Long Chen, Qichao Zhang†
ECCV, 2026

MeanFuser: Fast One-Step Multi-Modal Trajectory Generation and Adaptive Reconstruction via MeanFlow for End-to-End Autonomous Driving

Junli Wang, Zheng Yinan, Liu Xueyi, **Zebin Xing**, Pengfei Li, Kun Ma, Hangjun Ye, Guang Chen, Guang Li, Long Chen, Zhongpu Xia, Qichao Zhang†
CVPR, 2026

World4Drive: End-to-End Autonomous Driving via Intention-aware Physical Latent World Model

Yupeng Zheng, Pengxuan Yang, **Zebin Xing**, Qichao Zhang†, Yuhang Zheng, Yinfeng Gao, Pengfei Li, Teng Zhang, Zhongpu Xia, Peng Jia, Dongbin Zhao
ICCV, 2025

UncAD: Towards Safe End-to-end Autonomous Driving via Online Map Uncertainty

Pengxuan Yang, Yupeng Zheng, Qichao Zhang†, Kefei Zhu, **Zebin Xing**, Qiao Lin, Yunfu Liu, Zhiguo Su, Dongbin Zhao
ICRA, 2025

OccRWKV: Rethinking Efficient 3D Semantic Occupancy Prediction with Linear Complexity

Junming Wang, Wei* Yin, Xiaoxiao† Long, Xingyu Zhang, **Zebin Xing**, Xiaoyang Guo, Qian Zhang
ICRA, 2025

ComDrive: Comfort-Oriented End-to-End Autonomous Driving

Junming Wang, Xingyu Zhang, **Zebin Xing**, Songen Gu, Xiaoyang Guo, Yang Hu, Ziyang Song, Qian Zhang, Xiaoxiao Long, Wei Yin†
IROS, 2025

Challenge

ICCV Autonomous Grand Challenge 2025

CASIA | **Top contributor of the team**

Jun 2025 - Sep 2025

Third Place in NAVSIM v2 End-to-End Driving Challenge Track